

Banff National Park Activity Trends

Activity Trends 2020-25

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1 Land Acknowledgement

Biodiversity Pathways respectfully acknowledges that our work takes place on the territories of Treaties 6, 7, 8, and the Métis homeland, traditional and ancestral lands of First Nations and Métis Peoples, whose histories, languages, and cultures are directly linked to the biodiversity that we monitor.

We acknowledge the traditional teachings of the lands that we work on, and that reciprocal, meaningful, and respectful relationships with Indigenous peoples make our work possible. We are deeply grateful for their stewardship of these lands, and we are committed to supporting Indigenous-led monitoring programs, while learning Indigenous ways of knowing, being, and doing.

2 Introduction

2.1 Overview of NABat and the NNW Bat Hub

North American Bat Monitoring Program (NABat) is a large-scale coordinated effort to monitor bat species to address gaps in knowledge and lack of long-term studies across North America (Loeb et al. 2015). The program is administered by the US Geological Survey (USGS) and implemented by the North by Northwest Bat (NNW) Hub in Alberta, British Columbia, and S.E Alaska.

NNW Bat Hub was established in 2024 with collaboration from Government of Alberta, Government of British Columbia, and Government of Alaska. The hub was formed by combining the previous Alberta Hub and the BC and southeast Alaska Hub. The implementation of the program within Alberta has key support from Parks Canada, government of Alberta staff, other wildlife biologists, Indigenous Nations, community members and naturalists across the province.

2.1.1 NABat Monitoring in Banff National Park

Banff National Park has been conducting NABat monitoring within one NABat grid cell (Grid Cell ID: 148842) since 2020. Monitoring efforts include three long-term stationary sites and a driving transect. In addition, 10 supplementary sites were surveyed in 2022 and 2023 to develop a bat species inventory for the park. All data from 2020 to 2023 were processed using automated identification software. In 2024 and 2025 a subset of the data were also manually verified following the established NABat protocols ((Reichert et al. 2018a)).

2.2 Threats to bat populations in Alberta

Bat populations in Alberta face a wide range of threats including forestry practices, anthropogenic expansion, and climate change among others. Of these threats, White-Nose Syndrome and wind energy development have emerged as two of the highest-priority concerns.

2.2.1 White-Nose Syndrome

White-Nose Syndrome (WNS) is a deadly fungal disease caused by the fungus *Pseudogymnoascus destructans* (*Pd*) that affects hibernating bats. As of 2026, WNS has been confirmed in nine Canadian provinces including Alberta (Figure 1: CWHC-RCSF (2024)). Since *Pd* was first confirmed in AB in 2022, the fungus and the ensuing disease have spread through the province at a rapid pace and has now been detected at the provinces largest known hibernacula, Cadamun cave (Wilkinson, pers. comm.) Continued monitoring of populations in AB is critical as *Pd* and WNS continue to spread throughout the province.

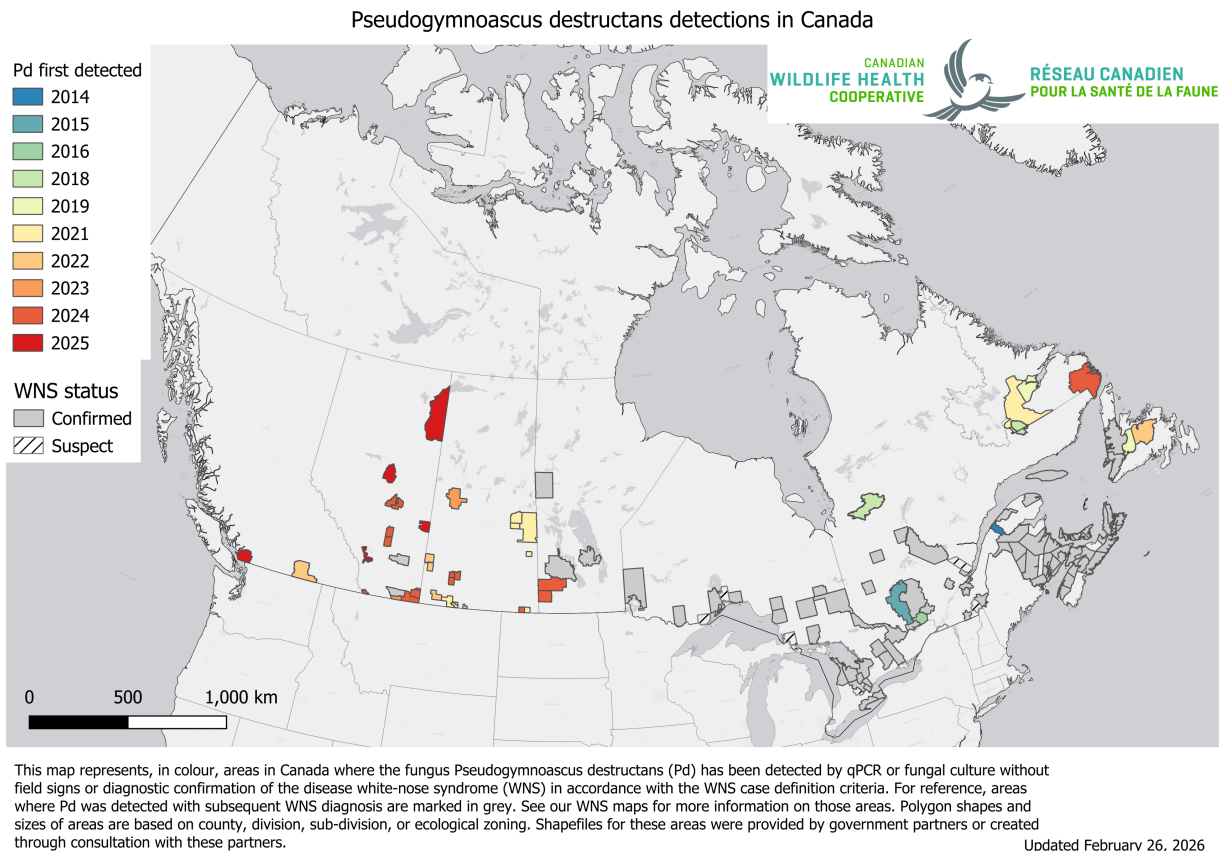


Figure 1: White-nose syndrome (WNS) and *Pseudogymnoascus destructans* (*Pd*) occurrence in Canada. Source: CWHC-RCSF (2026)

2.2.2 Wind Energy Development

Wind energy farms have been linked to high mortality rates across all bat species in Canada, with long distance migratory species accounting for 73% of recorded fatalities (Zimmerling and Francis 2016). In Alberta alone, one estimate predicts an annual mortality rate 10.9 bats per turbine (Zimmerling and Francis 2016). There are currently 1,778 turbines active in the province across 49 different projects with the most turbines found in the grassland ecosystem (“Canadian Wind Turbine Database” (2025) ; Figure 2). Using Zimmerling and Francis (2016) estimate, Alberta is estimated to be losing 19,380 bats per year to wind turbine collisions. As of 2023, COSEWIC designated all

three of Alberta's long distance migratory species as endangered due in part to fatalities at wind turbine facilities (Status of Endangered Wildlife in Canada 2023).



Figure 2: Current wind turbines in Alberta based on the Canadian Wind Turbine Database (Source: “Canadian Wind Turbine Database” (2025))

2.3 Acoustic Monitoring of Bats

Acoustic recording is a cost-effective method for monitoring bat populations across large geographic and temporal scales. However, several limitations affect the conclusions that can be drawn from these data:

1. **Imperfect species identification** — Bats rely on echolocation for hunting and moving around, and species with similar ecology and physiology often produce overlapping calls. While detectors are deployed to maximize recordings of diagnostic Search Phase calls, species identifiability will always vary depending on the local species assemblage.

2. **Variable detectability** — Bat species are not equally detectable acoustically. Larger, wide-ranging species produce louder calls that travel farther, while smaller species with quieter calls or specialized habitat use are less reliably recorded. As a result, trend estimates cannot be produced for all species using acoustic methods alone.
3. **Data variability** — Many factors influence recording quality and bat activity, including wind speed, humidity, and local habitat. Additionally, stationary detectors cannot account for individual recapture, meaning acoustic data can only inform relative activity and occupancy.
4. **Processing limitations** — Auto-ID software enables efficient processing of large datasets but has known high misclassification rates for bats, making manual verification necessary to ensure accurate results.

3 Methods

3.1 2025 Deployments

3.1.1 Stationary Surveys

Banff biologists deployed three stationary bat detectors in strategically selected bat habitat in 3 distinct quadrants within the NABat grid cell (**?@fig-sites**). Stationary sites were deployed from June 24 until July 3, 2025. Deployment set up followed the standards set by Loeb et al. (2015).

3.1.2 Transect Surveys

The driving transect protocol closely follows that established by Loeb et al. (2015). Two replicates of a mobile transect are driven at 30 – 35km/h with a bat detector on the roof to record bats along the road. Transects are conducted on the nights of June 24 and July 1 2025 (**?@fig-sites**).

3.2 Manual Verification

Recordings from 2020 were processed using an automated classifier originally developed by the Government of Alberta called Alberta eBat. Recordings from 2022 and 2023 were classified using Kaleidoscope’s Bats of North America 5.4.0 classifier. These automatic classifiers were used to assign species identifications and exclude non-bat recordings.

Full-spectrum recordings from the 2024 and 2025 sampling period were processed using two automatic classifiers: Kaleidoscope’s Bats of North America 5.4.0 classifier and Sonobat 3.0’s Northwestern British Columbia classifier. Based on documented species ranges and prior detection data (Olson, n.d.), manual identification efforts focused on seven species: Big Brown Bats (*Eptesicus fuscus*), Eastern Red Bats (*Lasiurus borealis*), Silver-haired Bats (*Lasionycteris noctivagans*), Hoary Bat (*Lasiurus cinereus*), Little Brown Bats (*Myotis lucifugus*), Long-legged Myotis (*Myotis volans*) and Long-eared Myotis (*Myotis evotis*).

The analysis workflow followed processing standards established by the North American Bat Monitoring Program (NABat) (Reichert et al. 2018b). Only recordings assigned a species classification by either Kaleidoscope or Sonobat were selected for manual verification. For stationary acoustic

monitoring, recordings were manually vetted until at least one recording per species per site per night was confidently identified. For mobile transects, all recordings with automated species classifications were manually vetted. Species identifications were validated against reference call parameters described by Slough et al. (2022), Solick (2022), and Szewczak (2018), adhering to NABat manual vetting standards (Reichert et al. 2018b).

All the recordings and results can be found on Widltrax under the Banff National Park Organization, in the project called [Bat Community - Long Term Monitoring - NABat - 2020-2025](#).

3.3 Statistical Trends

Trends estimates used only autoID results unless the manual verification results indicated a bat recordings was actually noise. This was done to maintain consistency across the 5 years of study, since manual verification only started in 2024.

3.3.1 Data processing

Species presence and nightly activity were calculated from a site-night-species table constructed for all sampled nights. Sampling effort was defined as each unique site-night with acoustic recordings. Weather covariates were summarized for each sampling night and included mean temperature, mean relative humidity, total precipitation, and mean wind speed. Moon covariates were calculated for each site-night and included moon illumination and an indicator of moon position. Environmental covariates were standardized prior to modeling to aid convergence and comparability; the year term was left on its original scale so that trend estimates represent per-year change. Mean nightly passes included all recordings with species or couplet labels, with couplets split and counted toward both species.

For trend analyses, Little Brown Bat (*Myotis lucifugus*), Northern Myotis (*M. spetentrionalis*), and Long-legged Myotis (*M. volans*) were pooled into a single Myotis group (40KMyo) due to high similarity in their calls and a correspondingly high potential for autoID misclassification. Nonetheless, most recordings in this group were likely MYLU.

3.3.2 Statistical analysis

Statistical trend analyses were conducted for Big Brown Bat (*Eptesicus fuscus*), Eastern Red Bat (*Lasiurus borealis*), Hoary Bat (*Lasiurus cinereus*), Silver-haired Bat (*Lasiyonicteris noctivagas*), and 40KMYO (Myotis grouping). Models were fit to nightly detection counts using negative binomial generalized linear mixed models (GLMMs) with a log link. Sites were included as a random intercept. Candidate model structures included year, Julian date, quadratic Julian date, and species-specific covariates. Trend analyses are presented only for species confirmed to be present at the sites through manual verification and for which models fully converged.

Table 1: Monitoring effort and bat species detections in the Banff (2020–2025). Mean nightly recordings ($\pm$ SD) represent total bat detections per night summed across species. Species detections indicate presence (X) within each year and quadrant based on combined automated and manually reviewed identifications. 2024 and 2025 are bolded to highlight that these were the only years where manual verification occurred.

Year	Site	Nights	Mean Nightly Bat Recordings $\pm$ SD	EPFU	LABO	LACI	I
2020	Fenlands	18	124.0 $\pm$ 146.1	X	X	X	
2021	Fenlands	16	236.2 $\pm$ 330.7	X	X	X	
2021	Golf Course	15	506.3 $\pm$ 201.5	X	X	X	
2021	Upper Hotsprings	17	7.4 $\pm$ 9.8	X	X	X	
2022	Fenlands	14	334.9 $\pm$ 298.6	X	X	X	
2022	Upper Hotsprings	14	2.5 $\pm$ 2.6				X
2023	Golf Course	14	452.9 $\pm$ 246.5	X	X	X	
2023	Upper Hotsprings	14	5.4 $\pm$ 8.4	X	X	X	
2024	Fenlands	13	724.8 $\pm$ 468.4	X	X	X	
2024	Golf Course	13	375.6 $\pm$ 178.4	X	X	X	
2024	Upper Hotsprings	12	4.5 $\pm$ 6.0	X	X	X	
2025	Fenlands	9	34.6 $\pm$ 14.0	X	X	X	
2025	Golf Course	9	14.2 $\pm$ 8.3	X	X	X	
2025	Upper Hotsprings	9	4.1 $\pm$ 1.8	X	X	X	
2021	DrivingTransect	1	36.0 $\pm$ NA	X	X	X	
2022	DrivingTransect	3	57.7 $\pm$ 13.2	X	X	X	
2023	DrivingTransect	2	46.0 $\pm$ 7.1	X	X	X	
2024	DrivingTransect	3	12.7 $\pm$ 0.6	X	X		
2025	DrivingTransect	2	68.0 $\pm$ 8.5	X	X	X	

4 Results

Acoustic monitoring happened for a total of 187 detector nights across 3 stationary survey locations and 11 driving transects over six years (2020–2025) (Table 1). Survey effort was mostly even within sites and years with sites sampled for a mean of 13.4 nights per year (range: 9–r max(with(subset(nightly, !(Site ==`"DrivingTransect"`)), table(paste(year, Site))))nights per site-year). A total of 39706 bat detections were recorded across the three stationary sites and a total of 475 where collected during driving transects in the Banff NABat grid cell.

4.1 Species detected and mean nightly activity

Eight species were recorded across the study. Silver-haired bat (*Lasionycteris noctivagans*) and Little Brown Bat (*Myotis lucifugus*) were detected across every site and year. Northern myotis (*Myotis septentrionalis*) was by far the scarcest species, detected only at Fenlands in 2022 and 2024, and Golf Course in 2023 and 2024 (Table 1). The Little Brown Bat had the highest number of detections across all years (20354 ; nightly mean = 102.7979798). This was followed by Long-legged Myotis (2457 ; nightly mean = 12.4090909), and Eastern Red Bat (1348 ; nightly mean = 6.8080808).

Table 2: Species trend estimate model results. Need to explain what the estimate is telling us since this is in a log scale

Species	Covariates Used
EPFU	Nightly Mean Temperature + Nightly Mean Relative Humidity
LABO	Nightly Mean Temperature + Nightly Mean Relative Humidity
LACI	Nightly Mean Temperature + Nightly Precipitation + Moon Illumination Fraction
LANO	Nightly Mean Relative Humidity + Nightly Mean Wind Speed + Moon Above Horizon at Midnight (b
40KMyo	Nightly Mean Relative Humidity
MYEV	Nightly Precipitation + Moon Illumination Fraction + Nightly Mean Wind Speed + Nightly Mean Te

4.2 Species Trends

Models converged for six species/groups: Big Brown Bat (EPFU), Eastern Red Bat (LABO), Hoary Bat (LACI), Silver-haired Bat (LANO), Long-eared Myotis (MYEV) and the combined 40kHz Myotis group (40KMyo) (Figure 3).

5 Discussion

Data used for trends only uses autoID from Kaleidoscope, which has been shown to confuse Myotis spp. for LABO. so be careful here.

Results must be taken with caution because of small sample size, but overall can conclude that bat species at the grid cell are stable, except for LACI. This is no suprise seeing the increasing wind turbine development in the province and elsewhere.

How does this compare to the NABat abundance results for LABO, LACI and LANO in the whole region?

Keep in mind that in the winter of 2026 WNS has been confirmed across the Rockies and most importantly in Cadamun cave. This just points to the importance of continues monitoring in the region. This will likely be important to keep in mind and will mean the decrease of MYLU and MYSE

6 Acknowledgements

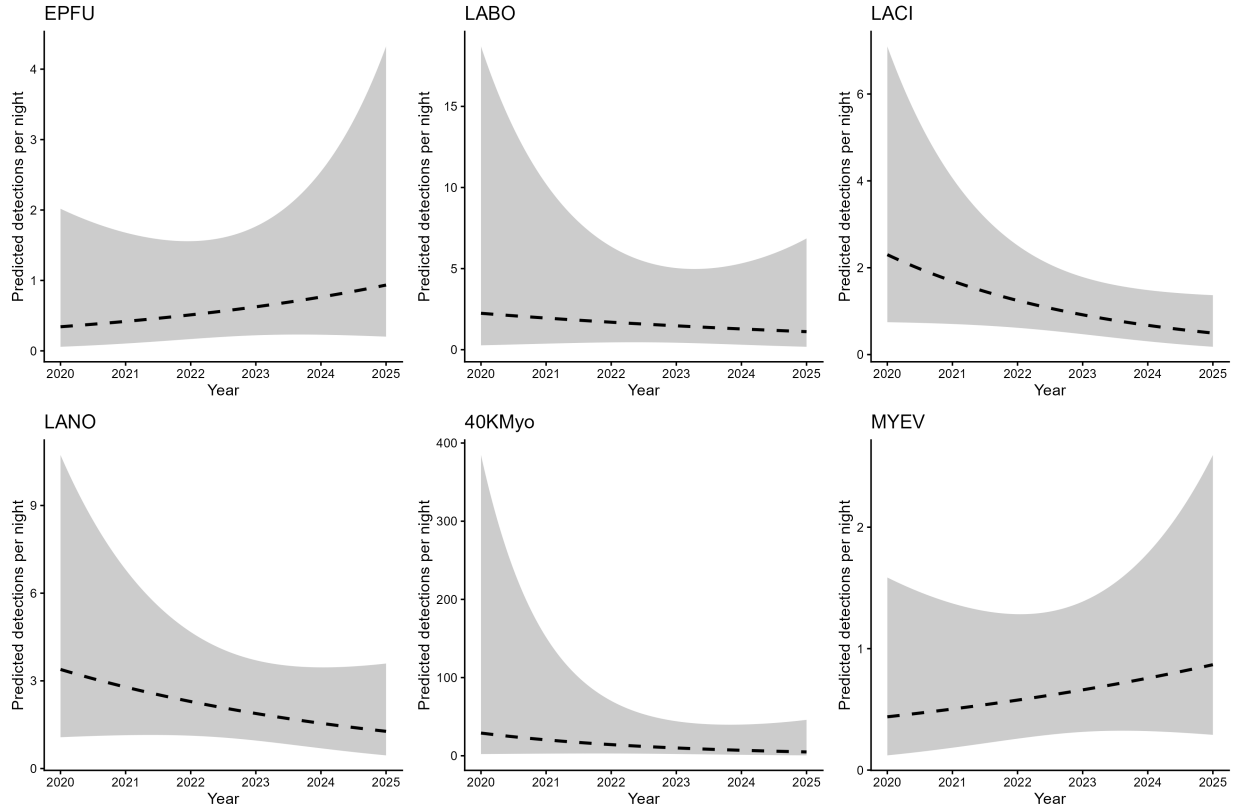


Figure 3: Species relative activity trends from 2020–2025 in the Banff NABat grid cell. Lines show trend estimates of nightly detections with 95% confidence intervals. Solid lines indicate significant trends ($p < 0.05$), and dashed lines indicate non-significant trends ($p > 0.05$). Models are based on nightly acoustic detections and fitted using negative binomial mixed-effects models.

7 Appendix A

Species codes and their definitions

CommonName	ScientificName	Code	Definition
Big Brown Bat	<i>Eptesicus fuscus</i>	EPFU	Calls that have diagnostic features identifying it as <i>Eptesicus fuscus</i>
Big Brown Bat / Silver-haired Bat	<i>Eptesicus fuscus</i> / <i>Lasionycteris noctivagans</i>	EPFULANO	Calls that could be attributed to either <i>Eptesicus fuscus</i> or <i>Lasionycteris noctivagans</i>
Eastern Red Bat	<i>Lasiurus borealis</i>	LABO	Calls that have diagnostic features identifying it as <i>Lasiurus borealis</i>
Eastern Red Bat / Little Brown Myotis	<i>Lasiurus borealis</i> / <i>Myotis lucifugus</i>	LABOMYLU	Calls that could be attributed to either <i>Lasiurus borealis</i> or <i>Myotis lucifugus</i>
Hoary Bat	<i>Lasiurus cinereus</i>	LACI	Calls that have diagnostic features identifying it as <i>Lasiurus cinereus</i>
Hoary Bat / Silver-haired Bat	<i>Lasiurus cinereus</i> / <i>Lasionycteris noctivagans</i>	LACILANO	Calls that could be attributed to either <i>Lasiurus cinereus</i> or <i>Lasionycteris noctivagans</i>
Silver-haired Bat	<i>Lasionycteris noctivagans</i>	LANO	Calls that have diagnostic features identifying it as <i>Lasionycteris noctivagans</i>
Little Brown Myotis	<i>Myotis lucifugus</i>	MYLU	Calls that have diagnostic features identifying it as <i>Myotis lucifugus</i>
Little Brown Myotis / Northern Myotis	<i>Myotis lucifugus</i> / <i>Myotis septentrionalis</i>	MYLUMYSE	Calls that could be attributed to either <i>Myotis lucifugus</i> or <i>Myotis septentrionalis</i>
Northern Myotis	<i>Myotis septentrionalis</i>	MYSE	Calls that have diagnostic features identifying it as <i>Myotis septentrionalis</i>
40kHz Frequency Myotis		40kMyo	Various species of <i>Myotis</i> that have a characteristic frequency in the range of 35-40kHz
High Frequency Bat		HIGH FRE-QUENCY	Various species with pulses having a characteristic frequency higher than ~35kHz
Low Frequency Bat		LOW FRE-QUENCY	Various species with pulses having a characteristic frequency lower than ~30kHz
Unknown Bat		NoID	Bat calls but no grouping category applies
Noise		Noise	No bat recorded

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